Silvia Casola

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Personal details:
- Date of birth: 08 July 1993
- Citizenship: Italian

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Research Interests

Natural Language Generation, Evaluation, Human Label Variation, Summarization, Simplification, Natural Language Processing, Deep Learning, Machine Learning.

Research Profile

I am a Marie Skłodowska-Curie Postdoctoral Fellow at Universitat Pompeu Fabra, Barcelona, hosted by Prof. Horacio Saggion. My project, GenEval, investigates the relationship between generation and evaluation in LLM judges, and uses these findings to predict their performance and create new metrics.
Previously, I was a post-doc researcher at the Ludwig Maximilian University of Munich, under the supervision of Prof. Barbara Plank, where I worked on NLG and evaluation, and at the University of Turin (PI: Prof. Valerio Basile), where I worked on perspective-aware NLP, disagreement, and human label variation (Amazon-Science-funded project).
I was awarded a Ph.D. cum laude by the University of Padua with a grant from Fondazione Bruno Kessler (Advisor: Alberto Lavelli); my main focus was on summarization and simplification, mostly in the patent domain. During my Ph.D., I visited Universitat Pompeu Fabra and interned remotely with Spotify Research and Huawei Research. I obtained a Laurea Magistrale (Master's Degree) with honors in Computer Engineering & Data Science from Politecnico di Torino and then worked as a research assistant at the same University.
My research interests lie in NLP, focusing on Natural Language Generation and evaluation.

Academic Appointments

- **Universitat Pompeu Fabra,** Barcelona, Spain *October 2026 – Present*
  Marie Skłodowska-Curie Postdoctoral Fellow.
  Project: GenEval: Linking Generation and Evaluation for Reliable NLG Assessment (HORIZON-MSCA-2025-PF).
  Advisor: Prof. Horacio Saggion

- **Ludwig Maximilian University of Munich,** Munich, Germany *September 2024 – August 2026*
  and **Munich Center for Machine Learning** (**MCML**)
  Post-doctoral Researcher & Lecturer
  Advisor: Prof. Barbara Plank
  I worked on NLG evaluation and variability (at the reference and language level). I also worked as a lecturer and supervised students.

- **University of Turin,** Turin, Italy *May 2023 – August 2024*
  Post-doctoral Researcher
  Advisor: Dr. Valerio Basile
  Project granted by Amazon Science
  I worked on perspectivism in Natural Language Processing. Our approaches exploit and analyze the disagreement among annotators rather than resolve it, e.g., through majority voting.

- **Fondazione Bruno Kessler,** Trento, Italy *February – May 2023*
  Research Fellow
  Advisor: Alberto Lavelli
  I worked on summarization and simplification with a focus on patents.

- **Politecnico di Torino,** Turin, Italy *January – October 2019*
  Research Assistant
  Advisor: Prof. Silvia Chiusano
  I was part of a team designing a recommendation system for health structures (CANP project).

Non-academic Work Experience

- **aequa-tech,** Remote *March – October 2024*
  NLP Research Scientist
  As part of the "Friends of aequa-tech" Fellowship program
  My work focused on using summarization in the news domain.

- **Spotify Research,** Remote *June – September 2023*
  NLP Research Scientist Intern
  My work focused on using LLMs to generate summaries of books.

- **Teleperformance Rome,** Remote *June – July 2023*
  Python instructor and trainer

- **Huawei Research Ireland,** Remote *July – December 2021*
  NLP Ph.D. Intern
  I developed a module to correct misspelled queries in the AppGallery, which has been deployed. Research-wise, I worked on ensuring factuality and minimizing hallucination in a sequence-to-sequence system for news summarization.

Education

- **University of Padua,** Padua, Italy *October 2019 – September 2023*
  Ph.D. in Brain, Mind and Computer Science
  cum laude and the additional label of Doctor Europaeus
  with a grant from Fondazione Bruno Kessler ($\sim$ 45K€)
  Advisor: Alberto Lavelli (FBK)
  Thesis: *Natural Language Processing for technology foresight – Summarization and simplification: the case of patents*

- **Politecnico di Torino,** Turin, Italy *2016 – 2018*
  Laurea Magistrale (2 years M.S. equivalent) in Computer Engineering (Data Science curriculum)
  Thesis: *Reducing waiting times and crowding in hospital emergency departments using Machine Learning*
  Supervisors: Prof. Ricard Gavaldà (UPC) and Prof. Silvia Chiusano (Politecnico di Torino)
  Grade: *110/110 cum laude*

- **Politecnico di Torino,** Turin, Italy *2013 – 2016*
  Laurea (3 years B.S. equivalent) in Computer Engineering.
  Grade: *102/110*

- **Robert Gordon University,** Aberdeen, UK *2012 – 2013*
  Certificate of Higher Education in Journalism

Other training activities

- **Universitat Pompeu Fabra,** Online *July 2026*
  Research Ethics and Personal Data Protection

- **University of Bari,** Bari, Italy *June 2024*
  AILC Lectures on Computational Linguistics 2024

- **Technische Universität München (TUM),** Munich, Germany *August 2021*
  1st Munich Legal Tech Summer School

- **University of Grenoble and Naver Labs Europe,** Online *January 2021*
  Advanced Language Processing Winter School (ALPS)

- **coursera.org**
  Deep Learning Specialization; by Prof. Andrew Ng (Stanford University) - Certificate
  Machine Learning Course; by Prof. Andrew Ng (Stanford University)
  Patent Law; by Prof. Polk Wagner (University of Pennsylvania)

Visiting Periods

- **Universitat Pompeu Fabra,** Barcelona, Spain *March – August 2022*
  Visiting Ph.D. Student
  I visited the TALN group under the supervision of Prof. Saggion. I collaborated with the group researchers and developed my interest in summarization and simplification.

- **Universitat Politècnica de Catalunya,** Barcelona, Spain *September 2017 – July 2018*
  Erasmus Program Master's student – MS in Artificial Intelligence, MS in Innovation and Research.
  I followed courses in the Machine Learning area and wrote and defended my Master's thesis.

Awards

- **Marie Skłodowska-Curie Actions European Postdoctoral Fellowship** *2026*
  Project: GenEval: Linking Generation and Evaluation for Reliable NLG Assessment (HORIZON-MSCA-2025-PF); acceptance rate: ~9%.
- **Marie Skłodowska-Curie Actions Seal of Excellence** *2025*
  Proposal: PERSEO - PERsonalized text Simplification and ElabOration (HORIZON-MSCA-2024-PF)
- **Outstanding Paper Award** *ACL 2024*
  MultiPICo: Multilingual Perspectivist Irony Corpus, Annual Meeting of the Association for Computational Linguistics

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Media coverage

- My project *GenEval: Linking Generation and Evaluation for Reliable NLG Assessment* was covered by the Universitat Pompeu Fabra (UPF) (link) and the TALN research group (link).

- Our survey to understand the use of GenAI tools in Italy was covered in the national newspaper "Corriere della Sera" in the article "Cosa fanno gli italiani con ChatGpt, Gemini & co? Un sondaggio per indagare l'uso (reale) dell'AI. E aiutare la ricerca" (link). The survey was also covered by Privacy network, TP24, and Bocconi news hub (link). The findings were further covered in Corriere della Sera in "Cosa facciamo (davvero) con ChatGpt, Gemini & co? I dati del primo studio sugli utenti italiani dei chatbot GenAI" (link).

- Our paper *MultiPICo: Multilingual Perspectivist Irony Corpus* was covered in the magazine "Otto: discorsi diretti" of the University of Torino in an interview with Prof. Valerio Basile (link).

Teaching

Post-Master courses

- **HumanAIze — Second-Level Master's Programme (Master di II livello)** *February 2025, 2026*
  Politecnico di Torino and University of Turin
  Lectures and laboratory sessions on NLP within the HumanAIze programme, a post-Master qualification open only to students who already hold a Master's degree. The course is aimed at providing humanities graduates with AI and digital competencies. Teaching load: 10 hours.

Master's Courses

- **Information Retrieval** *Summer Sem 2026*
  LMU Munich - Bachelor & Master in Computational Linguistics, Bachelor & Master in Computer Science; elective course for TUM students
  The course covers introductory concepts in IR and search engines. ~150 students. Teaching load: ~2h/week

- **Vertiefung der Grundlagen der Computerlinguistik** *Winter Sem 2025, Winter Sem 2026*
  LMU Munich - Master in Computational Linguistics
  The course covers NLP foundation topics (e.g., BPE tokenization, annotation, neural networks, word vectors, CNN, RNN, attention, transformers). Teaching load: ~2h/week

- **Trustworthy Data-centric AI** *Winter Sem 2025, Winter Sem 2026*
  LMU Munich - Master in Computational Linguistics
  The course covers advanced concepts in NLP, e.g., human annotation and crowdsourcing, silver data, FAIR principles, model behavioral testing, mechanistic interpretability, human and automatic evaluation, LLMs as judges. Teaching load: ~1h/week

- **NLG Evaluation** *Summer Sem 2025, 2026*
  LMU Munich - Master in Computational Linguistics
  I covered classes about automatic evaluation (standard metrics, LLMs as evaluators, data contamination, etc). Teaching load: ~2h/week

- **Summarization** *Summer Sem 2025, 2026*
  LMU Munich - Master in Computational Linguistics
  The course covered open topics in summarization and GenAI (hallucinations, agents, personalization, multimodality, etc) with a project-based student evaluation. The course was completely designed, prepared, and taught by me. Teaching load: ~2h/week

- **Simplification** *Winter Sem 2026*
  LMU Munich - Master in Computational Linguistics
  The course covered topics in simplification, with a project-based final evaluation. The course was completely designed, prepared, and taught by me. Teaching load: 2h/week

- **Summarization and Simplification** *Winter Sem 2025*
  LMU Munich - Master in Computational Linguistics
  The course covered topics in summarization and simplification, and the evaluation was project-based. The course was completely designed, prepared, and taught by me. Teaching load: 2h/week

Teaching Assistance

- **Programming (Python)** *Winter Sem 2020*
  University of Padua - Math BSc
  Lecturer, exerciser, and homework grader.

- **Computer Science with C** *Winter Sem 2017*
  Politecnico di Torino - BSc
  Exerciser and lab assistant.

Supervision

- ***Master's thesis supervision***
  - Xiang Shi, 2026 (LMU, advisor)
  - You Xu, 2026 (LMU, advisor)
  - Nils Grünefeld, 2025 (LMU, advisor)
  - Zhaochen Guo, 2025 (LMU, advisor)
- ***Bachelor thesis supervision***
  - Ziheng Zhou, 2026 (LMU, advisor)
  - Yueou Li, 2025 (LMU, advisor)

Academic service

- **Shared task organizer**
  - LeWiDi - Learning With Disagreement (co-located with the 4th Workshop on Perspectivist Approaches to NLP, EMNLP 2025)

- **Selection committee**
  - Invited selection committee member for the relAI scholarship for MSc students at TUM or LMU. The Konrad Zuse School of Excellence in Reliable AI (relAI) is a joint LMU and TUM program for AI-focused training.

- **Area Chair**
  - ACL Rolling Review (ARR) 2025, 2026
  - Generation, Evaluation & Metrics Workshop (GEM), 2026

- **Reviewer / PC member**
  - Journals
    - Computational Linguistics
    - IEEE Transactions on Affective Computing
    - Natural Language Engineering
    - AI & Ethics

  - Scientific Reports
  - Information Processing and Management
  - Neurocomputing
- Conferences
  - ACL Rolling Review (ARR)
  - International Natural Language Generation (INLG) Conference, 2026
  - Italian Conference on Computational Linguistics (CLiC-it), 2023, 2024, 2025, 2026
  - European Conference on Artificial Intelligence (ECAI), 2024
  - Empirical Methods in Natural Language Processing (EMNLP), 2023
- Workshops and other venues
  - ACL Student Research Workshop (ACL SRW), 2023, 2024, 2025, 2026
  - COLM Workshop on Bridging NLP and Public Opinion Research, 2025
  - RetroEval - Symposium on Natural Language Generation Evaluations, 2026
  - ACM/SIGAPP Symposium On Applied Computing, 2025, 2026
  - Generation, Evaluation & Metrics Workshop (GEM), 2022, 2023, 2025
  - Mind the AI Gap Workshop, 2025
  - NLPerspectives Workshop, 2024, 2025, 2026
  - Learning With Disagreement (LeWiDi), 2025, 2026
  - EACL Student Research Workshop (EACL SRW), 2024
  - Workshop on Natural Language for Artificial Intelligence @ AIxIA, 2024
  - European Symposium on Artificial Neural Networks (ESANN), 2020
  - Social Media Mining for Health Applications Workshop (#SMM4H), 2020

- **Mentor**
  ACL Student Research Workshop (2024)

- **Ph.D. course student representative** *February 2022 – February 2023*
  University of Padua

Invited talks and presentations

- ***STEEREVAL: Inference-time Interventions Strengthen Multilingual Generalization in Neural Summarization Metrics*** *2026*
  Empirical Methods in Natural Language Processing (EMNLP), Budapest
- ***Evaluating with Disagreement: Metrics for Soft and Perspectivist Evaluation of Categorical and Ordinal Judgments*** *2026*
  Dealing with Meaning Variation in NLP - 4th Yearly Workshop at Utrecht University, Online
- ***Evaluation Under Variation: References, Annotators, and Languages*** *2026*
  University of Turin, Turin
- ***Evaluation Under Variation: References, Annotators, and Languages*** *2026*
  Technische Universität München, Munich
- ***Evaluation Under Variation: References, Annotators, and Languages*** *2026*
  Dublin City University, Online
- ***References Matter: Investigating Reference Set Variation on Summarization Evaluation*** *2025*
  International Natural Language Generation (INLG) Conference, Hanoi
- ***Large Language Models in a polyphonic world*** *2024*
  Munich Center for Machine Learning Industry pitch talks, Munich RE, Munich
- ***MultiPICo: Multilingual Perspectivist Irony Corpus*** *2024*
  Annual Meeting of the Association for Computational Linguistics (ACL), Bangkok
- ***Confidence-based Ensembling of Perspective-aware Models*** *2024*
  Workshop on Perspectivist Approaches to NLP (NLPerspectives), LREC-COLING, Turin
- ***A perspectivist benchmark for training and evaluation*** *2024*
  2nd POPULUS Workshop, University of Turin
- ***Confidence-based Ensembling of Perspective-aware Models*** *2023*
  Empirical Methods in Natural Language Processing (EMNLP), Singapore

- ***Testing ChatGPT for Stability and Reasoning*** *2023*
  Italian Conference on Computational Linguistics (CLiC-it), Venice
- ***Creating a Silver Standard for Patent simplification*** *2023*
  Association for Computing Machinery's Special Interest Group on Information Retrieval Conference on Research and Development in Information Retrieval (SIGIR), Online
- ***Benchmarking Natural Language Processing Algorithms for Patent Summarization*** *2023*
  Workshop on Patent Text Mining and Semantic Technologies (PatentSemTech), SIGIR, Online
- ***Computers and Language*** *2023*
  Istituto di Istruzione Superiore Evangelista Torricelli, Milan
- ***What's in a (dataset's) name? The case of BigPatent*** *2022*
  Workshop on Generation, Evaluation & Metrics, (EMNLP), Online
- ***WITS: Wikipedia for Italian Text Summarization*** *2021*
  Italian Conference on Computational Linguistics (CLiC-it), Milan
- ***Natural Language Processing: an overview*** *2021*
  Machine Learning Course, the University of Padua (invited by Prof. F. Aiolli)
- ***FBK@SMM4H2020: RoBERTa for detecting medications on Twitter*** *2020*
  Social Media Mining for Health Applications Workshop & Shared Task (COLING), Online

Research projects

- ***GenEval: Linking Generation and Evaluation for Reliable NLG Assessment*** *2026 - 2028*
  Role: Marie Skłodowska-Curie Postdoctoral Fellow
  Funding: Horizon Europe (HORIZON-MSCA- 2025-PF)
  Advisor: Horacio Saggion (Universitat Pompeu Fabra).

- ***POPULUS: Multilingual Perspective-aware NLU*** *2023 - 2024*
  Role: Post-doc researcher
  Funding: Amazon Science
  PI: Valerio Basile (University of Turin).

- ***CANP: La Casa nel Parco*** *2019*
  Role: Research Assistant
  Funding: Regione Piemonte
  PI: Silvia Chiusano (Politecnico di Torino).

Other activities

CD: 50/50, co-founder
Non-profit association for promoting gender equality and diversity in STEM.
The association is part of Repubblica Digitale and sponsored by ASviS - Alleanza Italiana per lo Sviluppo Sostenibile. Some of my activities were:
- Liceo Scientifico Morgagni, Roma: course (PCTO) on Python and data literacy (2022, 40h)
- Istituto di Istruzione Superiore Evangelista Torricelli, Milano: Talk on Computers and language (2023)
- Istituto di Istruzione Superiore Piero Gobetti, Torino: course (PCTO) on AI and NLP (2024, 20hx3 classes)

Direction Under30, coordination, communication and organization
Coordination and organization of a project to select, discuss, and award theatrical projects in the context of the national DU30 Festival (2018–Present).

Technical Skills

Programming languages: Python (first language), SQL, C. Basics in R, C++, and Java.
Machine/Deep Learning stack: PyTorch; Numpy, Pandas, Matplotlib, Scipy, Scikit-learn, etc.; LangChain & Hugging Face ecosystems, NLTK, spaCy; Weights & Biases. Experience with large-scale GPU clusters (slurm).

Languages

Italian (native)
English (full professional proficiency) - C2 Certificate (10/2020)
Spanish (conversational proficiency)

List of Publications

- **Casola***, Shim*, Körner, Mao, Plank (2026): STEEREVAL: Inference-time Interventions Strengthen Multilingual Generalization in Neural Summarization Metrics, Empirical Methods in Natural Language Processing (EMNLP) 2026.
- Rizzi, **Casola**, Fersini, Leonardelli, Plank, Poesio (2026): Evaluating with Disagreement: Metrics for Soft and Perspectivist Evaluation of Categorical and Ordinal Judgments, Workshop on Pluralistic AI & NLP: Diversity-aware, Sociotechnical, Responsible Alignment (PANDORA), EMNLP 2026.
- Nissim, ..., **Casola**, ..., Zugarini (2026): Challenging the Abilities of Large Language Models in Italian: a Community Initiative, Italian Journal of Computational Linguistics.
- Longo, **Casola**, Martignetti, Noto, Stranisci (2026): The Teacher, the Slides, the Classroom, Italian Conference on Computational Linguistics (CLiC-it) 2026.
- Savoldi, Attanasio, Gorodetskaya, Marchiori Manerba, Bassignana, **Casola**, Negri, Caselli, Bentivogli, Ramponi, Muti, Balbo, Nozza (2026): Generative AI Practices, Literacy, and Divides: An Empirical Analysis in the Italian Context, arXiv.
- Zouhar, ..., **Casola**, ..., Alyafeai (2026): The Last Translation Benchmark, arXiv.
- Lo*, **Casola***, Sezerer, Basile, Sansonetti, Uva, Bernardi (2025): PERSEVAL: A Framework for Perspectivist Classification Evaluation, Empirical Methods in Natural Language Processing (EMNLP) 2025.
- Du, Mondorf, **Casola**, Yao, Litschko, Plank (2025): Reason to Rote: Rethinking Memorization in Reasoning, Empirical Methods in Natural Language Processing (EMNLP) 2025.
- **Casola***, Liu*, Peng*, Kraus , Gatt, Plank (2025): References Matter: Investigating the Impact of Reference Set Variation on Summarization Evaluation, International Natural Language Generation (INLG) Conference 2025.
- Leonardelli, **Casola**, Peng, Rizzi, Basile, Fersini, Frassinelli, Jang, Pavlovic, Plank, Poesio (2025): LeWiDi-2025 at NLPerspectives: The Third Edition of the Learning with Disagreements Shared Task, Workshop on Perspectivist Approaches to NLP (NLPerspectives), EMNLP 2025.
- Balestrucci*, **Casola***, Lo*, Basile, Mazzei (2024): I'm sure you're a real scholar yourself: Exploring Ironic Content Generation by Large Language Models, Empirical Methods in Natural Language Processing Findings (EMNLP Findings) 2024.
- **Casola***, Frenda*, Lo*, Sezerer, Uva, Basile, Bosco, Pedrani, Rubagotti, Patti, Bernardi (2024): MultiPICo: Multilingual Perspectivist Irony Corpus, Annual Meeting of the Association for Computational Linguistics (ACL) 2024 - **ACL 2024 Outstanding Paper**
- Bourgeade, **Casola**, Wizani, Bosco (2024): Data Augmentation through Back-Translation for Stereotypes and Irony Detection, Italian Conference on Computational Linguistics (CLiC-it) 2024.
- Basile, **Casola**, Frenda, Lo (2024): PERSEID-Perspectivist Irony Detection: A CALAMITA Challenge, Italian Conference on Computational Linguistics (CLiC-it) 2024.
- Frenda, Piergentili, Savoldi, Madeddu, Rosola, **Casola**, Ferrando, Patti, Negri, Bentivogli (2024): GFG-Gender-Fair Generation: A CALAMITA Challenge, Italian Conference on Computational Linguistics (CLiC-it) 2024.
- **Casola**, Lo, Basile, Frenda, Cignarella, Patti, Bosco (2023): Confidence-based Ensembling of Perspective-aware Models. Empirical Methods in Natural Language Processing (EMNLP) 2023
- **Casola**, Lavelli, Saggion (2023): Creating a Silver Standard for Patent simplification. Association for Computing Machinery's Special Interest Group on Information Retrieval Conference on Research and Development in Information Retrieval (SIGIR) 2023

- **Casola**, Labruna, Lavelli, Magnini (2023): Testing ChatGPT for Stability and Reasoning: A Case Study Using Italian Medical Specialty Tests. Italian Conference on Computational Linguistics (CLiC-it) 2023
- Frenda, Lo, **Casola**, Scarlini, Marco, Basile, Bernardi (2023): Does anyone see the irony here? Analysis of perspective-aware model predictions in irony detection. Workshop on Perspectivist Approaches to NLP (NLPerspectives), ECAI 2023
- **Casola**, Lavelli (2023): Benchmarking Natural Language Processing Algorithms for Patent Summarization. 3rd Workshop on Patent Text Mining and Semantic Technologies, SIGIR 2023
- **Casola**, Lavelli, Saggion (2022): What's in a (dataset's) name? The case of BigPatent. Workshop on Generation, Evaluation & Metrics (GEM), EMNLP 2022
- Obonyo, **Casola**, Saggion (2022): Exploring the limits of a base BART for multi-document summarization in the medical domain. Workshop on Scholarly Document Processing, COLING 2022
- **Casola**, Lavelli (2022): Summarization, Simplification, and Generation: The Case of Patents. Expert Systems with Applications
- **Casola**, Lauriola, Lavelli (2022): Pre-Trained Transformers: An Empirical Comparison. Machine Learning with Applications
- **Casola**, Lavelli (2021): WITS: Wikipedia for Italian Text Summarization. Italian Conference on Computational Linguistics (CLiC-it) 2021
- Louvan, **Casola**, Magnini (2021): Investigating Continued pretraining for Zero-Shot Cross-Lingual Spoken Language Understanding. Italian Conference on Computational Linguistics (CLiC-it) 2021
- **Casola**, Lavelli (2020): FBK@SMM4H2020: RoBERTa for detecting medications on Twitter. Social Media Mining for Health Applications Workshop & Shared Task, COLING 2020
- Bassi, **Casola**, Mancinelli, Lai, Salcuni (2020): Internalizing-Externalizing Symptoms as Predictors of Problematic Smartphone Use among Adolescents: A Machine Learning approach. International Congress of Clinical and Health Psychology in Children and Adolescents.
- Bazzano, Montuschi, Lamberti, Paravati, **Casola**, Ceròn, Londoño, Tanese (2017): Mental Workload Assessment for UAV Traffic Control Using Consumer-Grade BCI Equipment. Intelligent Human Computer Interaction (IHCI) Conference 2017.